Annexure 2

# SOFTWARE ENGINEERING

**Course Code: CSE 605** **Credit Units: 03**
**Total Hours: 45**

**Course Objective:**
The basic objective of Software Engineering is to develop methods and procedures for software development that can scale up for large systems and that can be used to consistently produce high-quality software at low cost and with a small cycle time. Software Engineering is the systematic approach to the development, operation, maintenance, and retirement of software.
The course provides a thorough introduction to the fundamentals principles of software engineering. The organization broadly is based on the classical analysis-design-implementation framework.

**Course Contents:**

**Module I: Introduction: (9 Hours)**
Software life cycle models: Waterfall, Prototype, Evolutionary and Spiral models, Overview of Quality Standards like ISO 9001, SEI-CMM.

**Module II: Software Metrics and Project Planning: (9 Hours)**
Size Metrics like LOC, Token Count, Function Count, Design Metrics, Data Structure Metrics, Information Flow Metrics. Cost estimation, static, Single and multivariate models, COCOMO model, Putnam Resource Allocation Model, Risk management.

**Module III: Software Requirement Analysis, design and coding: (10 Hours)**
Problem Analysis, Software Requirement and Specifications, Behavioural and non-behavioural requirements, Software Prototyping Cohesion & Coupling, Classification of Cohesiveness & Coupling, Function Oriented Design, Object Oriented Design, User Interface Design Top-down and bottom-up Structured programming, Information hiding.

**Module IV: Software Reliability, Testing and Maintenance: (11 Hours)**
Failure and Faults, Reliability Models: Basic Model, Logarithmic Poisson Model, Software process, Functional testing: Boundary value analysis, Equivalence class testing, Decision table testing, Cause effect graphing, Structural testing: path testing, Data flow and mutation testing, unit testing, integration and system testing, Debugging, Testing Tools, & Standards.Management of maintenance, Maintenance Process, Maintenance Models, Reverse Engineering, Software RE-engineering.

**Module V: UML: (6 Hours)**
Introduction to UML,Use Case Diagrams, Class Diagram: State Diagram in UMLActivity Diagram in UMLSequence Diagram in UMLCollaboration Diagram in UML.

## Course Outcomes:

- Ability to use the modeling approaches for the designing of software.
- Ability to use the testing tools and designing of test cases for testing.
- Ability to use the Unified modeling language (UML) for the designing of software product.

**Examination Scheme:**

| Components | A | CT | S/V/Q/HA | ESE |
|---|---|---|---|---|
| Weightage (%) | 5 | 15 | 10 | 70 |

A: Attendance, CT: Class Test,:, S/V/Q/HA: Seminar/Viva/Quiz/ Home Assignment, ESE: End Semester Examination;

## Text & References:

***Text:***


- K. K. Aggarwal & Yogesh Singh, "Software Engineering", 2nd Ed, New Age International, 2005.
- R. S. Pressman, "Software Engineering – A practitioner's approach", 5th Ed., McGraw Hill Int. Ed., 2001.


***References:***


- R. Fairley, "Software Engineering Concepts", Tata McGraw Hill, 1997.
- P. Jalote, "An Integrated approach to Software Engineering", Narosa, 1991.